

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I KANMANI KB ¹⁹²⁵²⁴³⁰⁶ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kanmani

1. Intelligent Agents in a Smart Home Automation System

Introduction:

A smart home automation system uses intelligent agents to control lighting, temperature & security, based on user behaviour & environmental conditions. It uses sensors to collect information and makes decisions to improve comfort.

Types of Intelligent Agents

1. Simple reflex agent

→ A simple reflex agent makes decisions based on the current situation using condition-action rules.

- Eg: Turns ON light when motion is detected.
- It is fast & simple.
- Cannot remember past situations or learn user preferences.

2. Model-Based Agent

→ A model-based agent maintains information about the environment and uses past & current data to make decisions.

- Eg: Remembers whether a room is occupied.
- Works well in changing environments.
- Requires more memory and processing.

3. Goal-Based Agent
- A goal-based agent selects actions to achieve specific goals.
- eg: Adjusts lighting and temperature to maintain comfort & security.
 - Helps achieve desired results.
 - Planning may take more time.

4. Utility-Based Agent
- A utility based agent selects action by balancing different factors, such as comfort, cost & energy consumption.
- eg: maintains a comfortable temperature while reducing electricity usage.
 - Balances multiple requirements.
 - Selecting the best action can be complex.

5. Learning Agent
- A learning agent improves its performance by learning from past experience & user preferences.
- eg: Learns the user's daily routine & adjusts lights automatically.
 - Provides personalized automation.
 - Requires data & privacy protection.

Search Algorithms for a Robot in an Unknown Maze

Introduction:

A robot placed in an unknown maze must find a path to the exit without prior knowledge of the layout. Uninformed search algorithms, such as Uniform Cost Search (UCS) & Iterative Deepening Search (IDS) help the robot explore possible paths & reach the goal.

1. Uniform Cost Search (UCS)

→ Uniform Cost Search expands the position with the lowest total path cost first. It uses a priority queue to select the next position to explore.

Working:

1. Start from the initial position.
2. Identify the available paths.
3. Calculate the total cost of reaching each position.
4. Explore the position with the lowest cost.
5. Continue until the exit is found.

Advantages:

- Finds the least-cost path when step costs are positive.

Disadvantage:

- Requires more memory.

Time & Space Complexity:

Both can be exponential in the worst case.

2. Iterative Deepening Search (IDS)

→ Iterative deepening search performs depth-limited searches repeatedly, increasing the depth limit each time until the unit is found.

Working:

1. Start searching with a depth limit of zero.
2. Explore all positions within that limit.
3. Increase the depth limit by one.
4. Repeat until the unit is reached.

Advantages:

- Uses less memory than BFS
- Complete when the branching factor is finite.

Disadvantage:

- Repeat some searches.
- Does not guarantee the least-cost path when movements cost differ.

Time complexity: $O(b^d)$

Space complexity: $O(bd)$

Here, b is the branching factor and d is the depth of the solution.